

Computer Networks (2024)

Homework 1 - Reliable Communication

Please read the sections on Go-Back-N and Selective repeat (3.4.3 and 3.4.4 in the 6th, 7th and 8th editions of the textbook).

Exercise 1 – TCP Congestion Control

Answer each of the following questions in one sentence. (For reference, use Figure 3.51, FSM description of TCP congestion control)

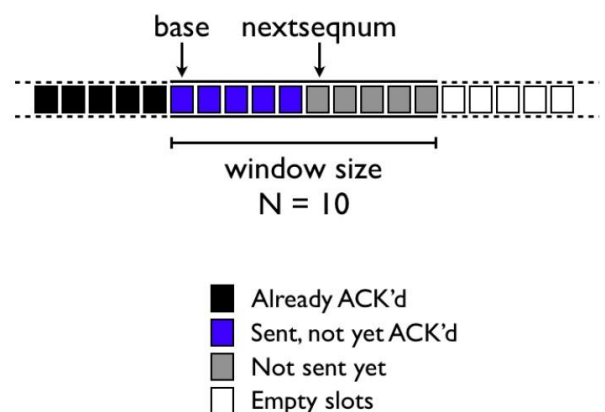
- 1) Why does TCP repeatedly increase the congestion window until it detects a problem (timeout or triple duplicate ACKs), then decrease it instead of finding an optimal maximum then staying there?
- 2) Why are there two distinct phases, one with an exponential increase in congestion window, and one with a linear increase.
- 3) During congestion avoidance, the congestion window increases linearly, but upon receiving triple duplicate acks, why does the congestion window halve instead of decreasing linearly.

Exercise 2 – Repeat

- a. Visit the Go-Back-N Java applet at this [website](#). Try it out and get familiar with how it works then, using the applet, answer the following questions:
 1. Have the source send five packets and the pause the animation before any of the five packets reach the destination. Then kill the second packet and resume the animation. Describe **what happens** and **why**.
 2. Repeat the experiment but now let the second packet reach the destination and kill the second acknowledgement. Describe again **what happens** and **why**.
- b. Visit the Selective Repeat Java applet at this [website](#) and repeat those questions.

Exercise 3 – Sliding Window

Using the figure above, answer a few questions related to the sliding window principle. The figure is imitated from the section on Go-back-n in the book.



- a. Draw the figure again showing where the **base**, **nextseqnum** are, and how the **box colors** change after each of the following events:
 1. One new packet has been sent from the transmitter, but an ACK has not been received.
 2. The ACK for the packet at the base is received by the transmitter.
 3. The transmitter receives an ACK for all the sent packets (assume all ACKs are received in order)

- b. How many bits would you need to indicate the sequence number in each packet with the following protocols? (Hint: Consider the window size.) Answer with formulas, and make sure to avoid ambiguity.
1. Go-Back-N
 2. Selective repeat